Assume by way of contradiction that L is context-free and let n be the value from the Pumping Lemma. Let $w = 0^n 1^n 2^n \in L$ and let u, v, x, y, z be strings such that $w = uvxyz$, $|vxy| \leq n$, and $|vy| \geq 1$. We can see that vxy cannot contain all three of 0s, 1s, and 2s since $|vxy| \leq n$ and the last 0 and first 2 in w are seperated by n 1s. If $vxy = 0^k$, then $1 \leq k \leq n$ since $|vy| \geq 1$ and $|vxy| \leq n$. Then, uv^2xy^2z contains more 0s than 1s and thus is not in the language. Similarly, if $vxy = 1^k$, then uv^2xy^2z contains more 1s than 0s and thus is not in the language. If $vxy = 2^k$, then uv^2xy^2z produces more 2s than 0s and 1s and thus is not in the language.

If vxy contains both 0s and 1s, then uv^0xy^0z contains an unequal number of 0s and 1s or more 2s than 0s or 1s and thus is not in the language. Finally, if vxy contains both 1s and 2s, then uv^2xy^2z either contains more 1s than 0s if we pump 1s or more 2s than 0s and 1s if we only pump 2s and is thus not in the language. In each case, we have a contradiction, and thus L cannot be context-free.